Set Operations

CS1010 Discrete Mathematics for Computer Science

Shibam Ghosh

August 13, 2026

Department of CSE, IIT Hyderabad

Set Operations

Union

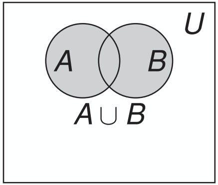
Definition

The **union** of sets A and B, denoted by $A \cup B$, is the set containing elements that are in A, or in B, or in both.

$$A \cup B = \{x \mid x \in A \vee x \in B\}$$

$\mathcal{Q}$. What is $\{1, 3, 9\} \cup \{3, 6, 9\}$?

$\mathcal{A}$. $\{1, 3, 6, 9\}$ (Notice that 3 and 9 are not written twice.)



Intersection

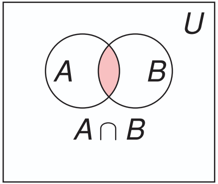
Definition

The **intersection** of sets A and B, denoted by $A \cap B$, is the set containing elements that are in both A and B.

$$A \cap B = \{x \mid x \in A \land x \in B\}$$

Disjoint Sets: If two sets have no elements in common (e.g., $A \cap B = \emptyset$)

- $\{2, 4, 6, 8, 10\} \cap \{1, 2, 3, 5, 7\} = \{2\}$
- $\{2, 4, 6, 8\} \cap \{1, 3, 5, 7, 9\} = \emptyset$



Theorem: For any finite sets A and B: $|A \cup B| = |A| + |B| - |A \cap B|$

Intersection

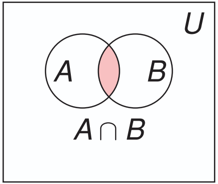
Definition

The **intersection** of sets A and B, denoted by $A \cap B$, is the set containing elements that are in both A and B.

$$A \cap B = \{x \mid x \in A \land x \in B\}$$

Disjoint Sets: If two sets have no elements in common (e.g., $A \cap B = \emptyset$)

- $\{2, 4, 6, 8, 10\} \cap \{1, 2, 3, 5, 7\} = \{2\}$
- $\{2, 4, 6, 8\} \cap \{1, 3, 5, 7, 9\} = \emptyset$



Theorem: For any finite sets A and B: $|A \cup B| = |A| + |B| - |A \cap B|$

Generalized Unions and Intersections

Definitions

Let $A_1, A_2, \ldots, A_n$ be an indexed collection of sets:

$$\bigcup_{i=1}^{n} A_i = A_1 \cup A_2 \cup \cdots \cup A_n \text{ and } \bigcap_{i=1}^{n} A_i = A_1 \cap A_2 \cap \cdots \cap A_n$$

$\mathcal{Q}$. For $i = 1, 2, 3, \ldots$ and let $A_i = [-i, i]$. Find $\bigcup_{i=1}^{\infty} A_i$ and $\bigcap_{i=1}^{\infty} A_i$.

$\mathcal{A}$. **Generalized Union:** As i grows, the intervals expand outward: $[-1, 1] \subset [-2, 2] \subset [-3, 3] \subset \ldots$ Eventually, the union covers all real numbers:

$$\bigcup_{i=1}^{\infty} A_i = (-\infty, \infty) = \mathbb{R}$$

$\mathcal{A}$. **Generalized Intersection:** $\bigcap_{i=1}^{\infty} A_i = A_1 = [-1, 1]$

Generalized Unions and Intersections

Definitions

Let $A_1, A_2, \ldots, A_n$ be an indexed collection of sets:

$$\bigcup_{i=1}^{n} A_i = A_1 \cup A_2 \cup \cdots \cup A_n \text{ and } \bigcap_{i=1}^{n} A_i = A_1 \cap A_2 \cap \cdots \cap A_n$$

$\mathcal{Q}$. For $i = 1, 2, 3, \ldots$ and let $A_i = [-i, i]$. Find $\bigcup_{i=1}^{\infty} A_i$ and $\bigcap_{i=1}^{\infty} A_i$.

$\mathcal{A}$. **Generalized Union:** As i grows, the intervals expand outward: $[-1, 1] \subset [-2, 2] \subset [-3, 3] \subset \ldots$ Eventually, the union covers all real numbers:

$$\bigcup_{i=1}^{\infty} A_i = (-\infty, \infty) = \mathbb{R}$$

$\mathcal{A}$. **Generalized Intersection:** $\bigcap_{i=1}^{\infty} A_i = A_1 = [-1, 1]$

Generalized Unions and Intersections

Definitions

Let $A_1, A_2, \ldots, A_n$ be an indexed collection of sets:

$$\bigcup_{i=1}^{n} A_i = A_1 \cup A_2 \cup \cdots \cup A_n \text{ and } \bigcap_{i=1}^{n} A_i = A_1 \cap A_2 \cap \cdots \cap A_n$$

$\mathcal{Q}$. For $i = 1, 2, 3, \ldots$ and let $A_i = [-i, i]$. Find $\bigcup_{i=1}^{\infty} A_i$ and $\bigcap_{i=1}^{\infty} A_i$.

$\mathcal{A}$. **Generalized Union:** As i grows, the intervals expand outward: $[-1, 1] \subset [-2, 2] \subset [-3, 3] \subset \ldots$ Eventually, the union covers all real numbers:

$$\bigcup_{i=1}^{\infty} A_i = (-\infty, \infty) = \mathbb{R}$$

$\mathcal{A}$. **Generalized Intersection:** $\bigcap_{i=1}^{\infty} A_i = A_1 = [-1, 1]$

Quick Check: Union and Intersection

Union asks: “Does it belong to at least one?”
Intersection asks: “Does it belong to all?”

Quick Check: Union and Intersection

Union asks: "Does it belong to at least one?"
Intersection asks: "Does it belong to all?"

$\mathcal{Q}$. Let $A = \{1, 2, 3, 4, 5\}$, and $B = \{4, 5, 6, 7\}$. Find $A \cup B$ and $A \cap B$.

Quick Check: Union and Intersection

Union asks: "Does it belong to at least one?"
Intersection asks: "Does it belong to all?"

$\mathcal{Q}$. Let $A = \{1, 2, 3, 4, 5\}$, and $B = \{4, 5, 6, 7\}$. Find $A \cup B$ and $A \cap B$.

$\mathcal{A}$. $A \cup B = \{1, 2, 3, 4, 5, 6, 7\}$, and $A \cap B = \{4, 5\}$.

Quick Check: Union and Intersection

Union asks: "Does it belong to at least one?"
Intersection asks: "Does it belong to all?"

$\mathcal{Q}$. Let $A = \{1, 2, 3, 4, 5\}$, and $B = \{4, 5, 6, 7\}$. Find $A \cup B$ and $A \cap B$.

$\mathcal{A}$. $A \cup B = \{1, 2, 3, 4, 5, 6, 7\}$, and $A \cap B = \{4, 5\}$.

$\mathcal{Q}$. Suppose $|A| = 20$, $|B| = 15$, $|A \cap B| = 6$. What is $|A \cup B|$?

Quick Check: Union and Intersection

Union asks: "Does it belong to at least one?"
Intersection asks: "Does it belong to all?"

$\mathcal{Q}$. Let $A = \{1, 2, 3, 4, 5\}$, and $B = \{4, 5, 6, 7\}$. Find $A \cup B$ and $A \cap B$.

$\mathcal{A}$. $A \cup B = \{1, 2, 3, 4, 5, 6, 7\}$, and $A \cap B = \{4, 5\}$.

$\mathcal{Q}$. Suppose $|A| = 20$, $|B| = 15$, $|A \cap B| = 6$. What is $|A \cup B|$?

$\mathcal{A}$. $|A \cup B| = |A| + |B| - |A \cap B| = 20 + 15 - 6 = 29$.

Quick Check: Union and Intersection

Union asks: "Does it belong to at least one?"
Intersection asks: "Does it belong to all?"

$\mathcal{Q}$. Let $A = \{1, 2, 3, 4, 5\}$, and $B = \{4, 5, 6, 7\}$. Find $A \cup B$ and $A \cap B$.

$\mathcal{A}$. $A \cup B = \{1, 2, 3, 4, 5, 6, 7\}$, and $A \cap B = \{4, 5\}$.

$\mathcal{Q}$. Suppose $|A| = 20$, $|B| = 15$, $|A \cap B| = 6$. What is $|A \cup B|$?

$\mathcal{A}$. $|A \cup B| = |A| + |B| - |A \cap B| = 20 + 15 - 6 = 29$.

$\mathcal{Q}$. If $A \cup B = A$, what can you conclude about A and B?

Quick Check: Union and Intersection

> *Union asks: "Does it belong to at least one?"*
> *Intersection asks: "Does it belong to all?"*

$\mathcal{Q}$. Let $A = \{1, 2, 3, 4, 5\}$, and $B = \{4, 5, 6, 7\}$. Find $A \cup B$ and $A \cap B$.

$\mathcal{A}$. $A \cup B = \{1, 2, 3, 4, 5, 6, 7\}$, and $A \cap B = \{4, 5\}$.

$\mathcal{Q}$. Suppose $|A| = 20, |B| = 15, |A \cap B| = 6$. What is $|A \cup B|$?

$\mathcal{A}$. $|A \cup B| = |A| + |B| - |A \cap B| = 20 + 15 - 6 = 29.$

$\mathcal{Q}$. If $A \cup B = A$, what can you conclude about A and B?

$\mathcal{A}$. $B \subseteq A$. Every element of B must already belong to A.

Quick Check: Union and Intersection

Union asks: "Does it belong to at least one?"
Intersection asks: "Does it belong to all?"

$\mathcal{Q}$. Let $A = \{1, 2, 3, 4, 5\}$, and $B = \{4, 5, 6, 7\}$. Find $A \cup B$ and $A \cap B$.

$\mathcal{A}$. $A \cup B = \{1, 2, 3, 4, 5, 6, 7\}$, and $A \cap B = \{4, 5\}$.

$\mathcal{Q}$. Suppose $|A| = 20$, $|B| = 15$, $|A \cap B| = 6$. What is $|A \cup B|$?

$\mathcal{A}$. $|A \cup B| = |A| + |B| - |A \cap B| = 20 + 15 - 6 = 29$.

$\mathcal{Q}$. If $A \cup B = A$, what can you conclude about A and B?

$\mathcal{A}$. $B \subseteq A$. Every element of B must already belong to A.

$\mathcal{Q}$. If $A \cap B = A$, what can you conclude about A and B?

Quick Check: Union and Intersection

Union asks: "Does it belong to at least one?"
Intersection asks: "Does it belong to all?"

$\mathcal{Q}$. Let $A = \{1, 2, 3, 4, 5\}$, and $B = \{4, 5, 6, 7\}$. Find $A \cup B$ and $A \cap B$.

$\mathcal{A}$. $A \cup B = \{1, 2, 3, 4, 5, 6, 7\}$, and $A \cap B = \{4, 5\}$.

$\mathcal{Q}$. Suppose $|A| = 20$, $|B| = 15$, $|A \cap B| = 6$. What is $|A \cup B|$?

$\mathcal{A}$. $|A \cup B| = |A| + |B| - |A \cap B| = 20 + 15 - 6 = 29$.

$\mathcal{Q}$. If $A \cup B = A$, what can you conclude about A and B?

$\mathcal{A}$. $B \subseteq A$. Every element of B must already belong to A.

$\mathcal{Q}$. If $A \cap B = A$, what can you conclude about A and B?

$\mathcal{A}$. $A \subseteq B$. Every element of A belongs to $A \cap B$, and therefore belongs to B.

Complement

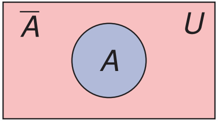
Definition

The **complement** of set A (with respect to U), denoted by $\overline{A}$ (or A^c), is the set of elements in U that are not in A.

$$\overline{A} = \{x \in U \mid x \notin A\} = U - A$$

$\mathcal{Q}.$ Let $U = \{0, 1, 2, \ldots, 9\}$. What is the complement of the set of even digits $E = \{x \in U \mid x \text{ is even}\}$?

$\mathcal{A}.$ $\overline{E} = \{1, 3, 5, 7, 9\}$ (the set of odd digits in U).



Difference

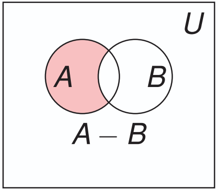
Definition

The **difference** of sets A and B, denoted by $A - B$, is the set containing elements of A that are not in B.

$$A - B = \{x \mid x \in A \wedge x \notin B\} = A \cap \overline{B}$$

Example:

- $A = \{\text{Red}, \text{Blue}, \text{Yellow}\}$, $B = \{\text{Red}, \text{Blue}, \text{Green}\}$
- $A - B = \{\text{Yellow}\}$ and $B - A = \{\text{Green}\}$
- In general, $A - B \neq B - A$



Summary of Basic Set Operations

Set operations provide a direct connection between **set theory and logic.**

Operation	Membership condition	Logical idea
$A \cup B$	$x \in A \vee x \in B$	OR
$A \cap B$	$x \in A \wedge x \in B$	AND
$\overline{A}$	$x \notin A$	NOT
$A - B$	$x \in A \wedge x \notin B$	AND + NOT

Quick Check: Set Operations

$\mathcal{Q}$. Let $U = \{1, 2, \ldots, 10\}$, $A = \{2, 4, 6, 8, 10\}$, $B = \{2, 3, 5, 7\}$. Find $\overline{A}$ and $A - B$.

Quick Check: Set Operations

$\mathcal{Q}$. Let $U = \{1, 2, \ldots, 10\}$, $A = \{2, 4, 6, 8, 10\}$, $B = \{2, 3, 5, 7\}$. Find $\overline{A}$ and $A - B$.

$\mathcal{A}$. $\overline{A} = \{1, 3, 5, 7, 9\}$, and $A - B = \{4, 6, 8, 10\}$.

Quick Check: Set Operations

$\mathcal{Q}$. Let $U = \{1, 2, \ldots, 10\}, A = \{2, 4, 6, 8, 10\}, B = \{2, 3, 5, 7\}$. Find $\overline{A}$ and $A - B$.

$\mathcal{A}$. $\overline{A} = \{1, 3, 5, 7, 9\}$, and $A - B = \{4, 6, 8, 10\}$.

$\mathcal{Q}$. Express $A - B$ using only intersection and complement.

Quick Check: Set Operations

$\mathcal{Q}$. Let $U = \{1, 2, \dots, 10\}$, $A = \{2, 4, 6, 8, 10\}$, $B = \{2, 3, 5, 7\}$. Find $\overline{A}$ and $A - B$.

$\mathcal{A}$. $\overline{A} = \{1, 3, 5, 7, 9\}$, and $A - B = \{4, 6, 8, 10\}$.

$\mathcal{Q}$. Express $A - B$ using only intersection and complement.

$\mathcal{A}$. $A - B = A \cap \overline{B}$.

Quick Check: Set Operations

$\mathcal{Q}$. Let $U = \{1, 2, \ldots, 10\}, A = \{2, 4, 6, 8, 10\}, B = \{2, 3, 5, 7\}$. Find $\overline{A}$ and $A - B$.

$\mathcal{A}$. $\overline{A} = \{1, 3, 5, 7, 9\}$, and $A - B = \{4, 6, 8, 10\}$.

$\mathcal{Q}$. Express $A - B$ using only intersection and complement.

$\mathcal{A}$. $A - B = A \cap \overline{B}$.

$\mathcal{Q}$. Let $A_i = \{i, i+1, i+2, \ldots\}, \qquad i = 1, 2, 3, \ldots$. What are $\bigcup_{i=1}^{\infty} A_i$ and $\bigcap_{i=1}^{\infty} A_i$?

Quick Check: Set Operations

$\mathcal{Q}$. Let $U = \{1, 2, \ldots, 10\}$, $A = \{2, 4, 6, 8, 10\}$, $B = \{2, 3, 5, 7\}$. Find $\overline{A}$ and $A - B$.

$\mathcal{A}$. $\overline{A} = \{1, 3, 5, 7, 9\}$, and $A - B = \{4, 6, 8, 10\}$.

$\mathcal{Q}$. Express $A - B$ using only intersection and complement.

$\mathcal{A}$. $A - B = A \cap \overline{B}$.

$\mathcal{Q}$. Let $A_i = \{i, i+1, i+2, \ldots\}, \qquad i = 1, 2, 3, \ldots$. What are $\bigcup_{i=1}^{\infty} A_i$ and $\bigcap_{i=1}^{\infty} A_i$?

$\mathcal{A}$. $\bigcup\limits_{i=1}^{\infty} A_i = \mathbb{Z}^+$ and $\bigcap\limits_{i=1}^{\infty} A_i = \emptyset$

Quick Check: Set Operations

$\mathcal{Q}$. Let $U = \{1, 2, \dots, 10\}$, $A = \{2, 4, 6, 8, 10\}$, $B = \{2, 3, 5, 7\}$. Find $\overline{A}$ and $A - B$.

$\mathcal{A}$. $\overline{A} = \{1, 3, 5, 7, 9\}$, and $A - B = \{4, 6, 8, 10\}$.

$\mathcal{Q}$. Express $A - B$ using only intersection and complement.

$\mathcal{A}$. $A - B = A \cap \overline{B}$.

$\mathcal{Q}$. Let $A_i = \{i, i+1, i+2, \dots\}, \qquad i = 1, 2, 3, \dots$. What are $\bigcup_{i=1}^{\infty} A_i$ and $\bigcap_{i=1}^{\infty} A_i$?

$\mathcal{A}$. $\displaystyle\bigcup_{i=1}^{\infty} A_i = \mathbb{Z}^+$ and $\displaystyle\bigcap_{i=1}^{\infty} A_i = \emptyset$

$\mathcal{Q}$. Suppose $A \cup B = A \cap B$. What can you conclude about A and B?

Quick Check: Set Operations

$\mathcal{Q}$. Let $U = \{1, 2, \ldots, 10\}, A = \{2, 4, 6, 8, 10\}, B = \{2, 3, 5, 7\}$. Find $\overline{A}$ and $A - B$.

$\mathcal{A}$. $\overline{A} = \{1, 3, 5, 7, 9\}$, and $A - B = \{4, 6, 8, 10\}$.

$\mathcal{Q}$. Express $A - B$ using only intersection and complement.

$\mathcal{A}$. $A - B = A \cap \overline{B}$.

$\mathcal{Q}$. Let $A_i = \{i, i+1, i+2, \ldots\}, \qquad i = 1, 2, 3, \ldots$. What are $\bigcup_{i=1}^{\infty} A_i$ and $\bigcap_{i=1}^{\infty} A_i$?

$\mathcal{A}$. $\bigcup\limits_{i=1}^{\infty} A_i = \mathbb{Z}^+$ and $\bigcap\limits_{i=1}^{\infty} A_i = \emptyset$

$\mathcal{Q}$. Suppose $A \cup B = A \cap B$. What can you conclude about A and B?

$\mathcal{A}$. $A = B$. Indeed, if an element belongs to A, then it belongs to $A \cup B$, and hence to $A \cap B$, so it must also belong to B. Similarly, $B \subseteq A$.

Quick Check: Set Operations

$\mathcal{Q}$. Let $U = \{1, 2, \ldots, 10\}$, $A = \{2, 4, 6, 8, 10\}$, $B = \{2, 3, 5, 7\}$. Find $\overline{A}$ and $A - B$.

$\mathcal{A}$. $\overline{A} = \{1, 3, 5, 7, 9\}$, and $A - B = \{4, 6, 8, 10\}$.

$\mathcal{Q}$. Express $A - B$ using only intersection and complement.

$\mathcal{A}$. $A - B = A \cap \overline{B}$.

$\mathcal{Q}$. Let $A_i = \{i, i+1, i+2, \ldots\}, \qquad i = 1, 2, 3, \ldots$. What are $\bigcup_{i=1}^{\infty} A_i$ and $\bigcap_{i=1}^{\infty} A_i$?

$\mathcal{A}$. $\bigcup\limits_{i=1}^{\infty} A_i = \mathbb{Z}^+$ and $\bigcap\limits_{i=1}^{\infty} A_i = \emptyset$

$\mathcal{Q}$. Suppose $A \cup B = A \cap B$. What can you conclude about A and B?

$\mathcal{A}$. $A = B$. Indeed, if an element belongs to A, then it belongs to $A \cup B$, and hence to $A \cap B$, so it must also belong to B. Similarly, $B \subseteq A$.

$\mathcal{Q}$. Can two non-empty sets satisfy $A \cap B = \emptyset$ and $A \cup B = \emptyset$?

Quick Check: Set Operations

$\mathcal{Q}$. Let $U = \{1, 2, \ldots, 10\}$, $A = \{2, 4, 6, 8, 10\}$, $B = \{2, 3, 5, 7\}$. Find $\overline{A}$ and $A - B$.

$\mathcal{A}$. $\overline{A} = \{1, 3, 5, 7, 9\}$, and $A - B = \{4, 6, 8, 10\}$.

$\mathcal{Q}$. Express $A - B$ using only intersection and complement.

$\mathcal{A}$. $A - B = A \cap \overline{B}$.

$\mathcal{Q}$. Let $A_i = \{i, i+1, i+2, \ldots\}$, $\quad i = 1, 2, 3, \ldots$. What are $\bigcup_{i=1}^{\infty} A_i$ and $\bigcap_{i=1}^{\infty} A_i$?

$\mathcal{A}$. $\displaystyle\bigcup_{i=1}^{\infty} A_i = \mathbb{Z}^+$ and $\displaystyle\bigcap_{i=1}^{\infty} A_i = \emptyset$

$\mathcal{Q}$. Suppose $A \cup B = A \cap B$. What can you conclude about A and B?

$\mathcal{A}$. $A = B$. Indeed, if an element belongs to A, then it belongs to $A \cup B$, and hence to $A \cap B$, so it must also belong to B. Similarly, $B \subseteq A$.

$\mathcal{Q}$. Can two non-empty sets satisfy $A \cap B = \emptyset$ and $A \cup B = \emptyset$?

$\mathcal{A}$. No. If $A \cup B = \emptyset$, then both A and B must be empty. Thus, $A = B = \emptyset$.

Set Identities: Core Laws (1)

Identity	Name
$A \cup \emptyset = A$ $A \cap U = A$	Identity Laws
$A \cup U = U$ $A \cap \emptyset = \emptyset$	Domination Laws
$A \cup A = A$ $A \cap A = A$	Idempotent Laws
$\overline{(\overline{A})} = A$	Complementation Law

Set Identities: Core Laws (2)

Identity	Name
$A \cup B = B \cup A$ $A \cap B = B \cap A$	Commutative Laws
$A \cup (B \cup C) = (A \cup B) \cup C$ $A \cap (B \cap C) = (A \cap B) \cap C$	Associative Laws
$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$	Distributive Laws

Set Identities: De Morgan's & Remaining Laws

Identity	Name
$\overline{A \cup B} = \overline{A} \cap \overline{B}$ $\overline{A \cap B} = \overline{A} \cup \overline{B}$	De Morgan's Laws
$A \cup (A \cap B) = A$ $A \cap (A \cup B) = A$	Absorption Laws
$A \cup \overline{A} = U$ $A \cap \overline{A} = \emptyset$	Complement Laws

Proof of De Morgan's Law

De Morgan's Law: $\boxed{\overline{A \cup B} = \overline{A} \cap \overline{B}}$

$\overline{A \cup B} = \{x \mid x \notin A \cup B\}$	by definition of complement
$= \{x \mid \neg(x \in A \cup B)\}$	by definition of $\notin$
$= \{x \mid \neg(x \in A \vee x \in B)\}$	by definition of union
$= \{x \mid \neg(x \in A) \wedge \neg(x \in B)\}$	by De Morgan's law for logic
$= \{x \mid x \notin A \wedge x \notin B\}$	by definition of $\notin$
$= \{x \mid x \in \overline{A} \wedge x \in \overline{B}\}$	by definition of complement
$= \{x \mid x \in \overline{A} \cap \overline{B}\}$	by definition of intersection
$= \overline{A} \cap \overline{B}$	by meaning of set-builder notation.

Proof of De Morgan's Law

De Morgan's Law: $\boxed{\overline{A \cup B} = \overline{A} \cap \overline{B}}$

$\overline{A \cup B} = \{x \mid x \notin A \cup B\}$	by definition of complement
$= \{x \mid \neg(x \in A \cup B)\}$	by definition of $\notin$
$= \{x \mid \neg(x \in A \vee x \in B)\}$	by definition of union
$= \{x \mid \neg(x \in A) \wedge \neg(x \in B)\}$	by De Morgan's law for logic
$= \{x \mid x \notin A \wedge x \notin B\}$	by definition of $\notin$
$= \{x \mid x \in \overline{A} \wedge x \in \overline{B}\}$	by definition of complement
$= \{x \mid x \in \overline{A} \cap \overline{B}\}$	by definition of intersection
$= \overline{A} \cap \overline{B}$	by meaning of set-builder notation.

Proof of De Morgan's Law

De Morgan's Law: $\boxed{\overline{A \cup B} = \overline{A} \cap \overline{B}}$

$\overline{A \cup B} = \{x \mid x \notin A \cup B\}$	by definition of complement
$= \{x \mid \neg(x \in A \cup B)\}$	by definition of $\notin$
$= \{x \mid \neg(x \in A \vee x \in B)\}$	by definition of union
$= \{x \mid \neg(x \in A) \wedge \neg(x \in B)\}$	by De Morgan's law for logic
$= \{x \mid x \notin A \wedge x \notin B\}$	by definition of $\notin$
$= \{x \mid x \in \overline{A} \wedge x \in \overline{B}\}$	by definition of complement
$= \{x \mid x \in \overline{A} \cap \overline{B}\}$	by definition of intersection
$= \overline{A} \cap \overline{B}$	by meaning of set-builder notation.

Proof of De Morgan's Law

De Morgan's Law: $\boxed{\overline{A \cup B} = \overline{A} \cap \overline{B}}$

$\overline{A \cup B} = \{x \mid x \notin A \cup B\}$	by definition of complement
$= \{x \mid \neg(x \in A \cup B)\}$	by definition of $\notin$
$= \{x \mid \neg(x \in A \vee x \in B)\}$	by definition of union
$= \{x \mid \neg(x \in A) \wedge \neg(x \in B)\}$	by De Morgan's law for logic
$= \{x \mid x \notin A \wedge x \notin B\}$	by definition of $\notin$
$= \{x \mid x \in \overline{A} \wedge x \in \overline{B}\}$	by definition of complement
$= \{x \mid x \in \overline{A} \cap \overline{B}\}$	by definition of intersection
$= \overline{A} \cap \overline{B}$	by meaning of set-builder notation.

Proof of De Morgan's Law

De Morgan's Law: $\boxed{\overline{A \cup B} = \overline{A} \cap \overline{B}}$

$$
\begin{aligned}
\overline{A \cup B} &= \{x \mid x \notin A \cup B\} && \text{by definition of complement} \\
&= \{x \mid \neg(x \in A \cup B)\} && \text{by definition of } \notin \\
&= \{x \mid \neg(x \in A \vee x \in B)\} && \text{by definition of union} \\
&= \{x \mid \neg(x \in A) \wedge \neg(x \in B)\} && \text{by De Morgan's law for logic} \\
&= \{x \mid x \notin A \wedge x \notin B\} && \text{by definition of } \notin \\
&= \{x \mid x \in \overline{A} \wedge x \in \overline{B}\} && \text{by definition of complement} \\
&= \{x \mid x \in \overline{A} \cap \overline{B}\} && \text{by definition of intersection} \\
&= \overline{A} \cap \overline{B} && \text{by meaning of set-builder notation.}
\end{aligned}
$$

Proof of De Morgan's Law

De Morgan's Law: $\boxed{\overline{A \cup B} = \overline{A} \cap \overline{B}}$

$$\overline{A \cup B} = \{x \mid x \notin A \cup B\}$$ by definition of complement

$$= \{x \mid \neg(x \in A \cup B)\}$$ by definition of $\notin$

$$= \{x \mid \neg(x \in A \vee x \in B)\}$$ by definition of union

$$= \{x \mid \neg(x \in A) \wedge \neg(x \in B)\}$$ by De Morgan's law for logic

$$= \{x \mid x \notin A \wedge x \notin B\}$$ by definition of $\notin$

$$= \{x \mid x \in \overline{A} \wedge x \in \overline{B}\}$$ by definition of complement

$$= \{x \mid x \in \overline{A} \cap \overline{B}\}$$ by definition of intersection

$$= \overline{A} \cap \overline{B}$$ by meaning of set-builder notation.

Proof of De Morgan's Law

De Morgan's Law: $\boxed{\overline{A \cup B} = \overline{A} \cap \overline{B}}$

$\overline{A \cup B} = \{x \mid x \notin A \cup B\}$	by definition of complement
$= \{x \mid \neg(x \in A \cup B)\}$	by definition of $\notin$
$= \{x \mid \neg(x \in A \vee x \in B)\}$	by definition of union
$= \{x \mid \neg(x \in A) \wedge \neg(x \in B)\}$	by De Morgan's law for logic
$= \{x \mid x \notin A \wedge x \notin B\}$	by definition of $\notin$
$= \{x \mid x \in \overline{A} \wedge x \in \overline{B}\}$	by definition of complement
$= \{x \mid x \in \overline{A} \cap \overline{B}\}$	by definition of intersection
$= \overline{A} \cap \overline{B}$	by meaning of set-builder notation.

Proof of De Morgan's Law

De Morgan's Law: $\boxed{\overline{A \cup B} = \overline{A} \cap \overline{B}}$

$$
\begin{aligned}
\overline{A \cup B} &= \{x \mid x \notin A \cup B\} && \text{by definition of complement} \\
&= \{x \mid \neg(x \in A \cup B)\} && \text{by definition of } \notin \\
&= \{x \mid \neg(x \in A \vee x \in B)\} && \text{by definition of union} \\
&= \{x \mid \neg(x \in A) \wedge \neg(x \in B)\} && \text{by De Morgan's law for logic} \\
&= \{x \mid x \notin A \wedge x \notin B\} && \text{by definition of } \notin \\
&= \{x \mid x \in \overline{A} \wedge x \in \overline{B}\} && \text{by definition of complement} \\
&= \{x \mid x \in \overline{A} \cap \overline{B}\} && \text{by definition of intersection} \\
&= \overline{A} \cap \overline{B} && \text{by meaning of set-builder notation.}
\end{aligned}
$$

Proof of the Absorption Law

Absorption Law: $\boxed{A \cup (A \cap B) = A}$

$$
\begin{aligned}
A \cup (A \cap B) &= \{x \mid x \in A \cup (A \cap B)\} && \text{by meaning of set-builder notation} \\
&= \{x \mid x \in A \vee x \in (A \cap B)\} && \text{by definition of union} \\
&= \{x \mid x \in A \vee (x \in A \wedge x \in B)\} && \text{by definition of intersection} \\
&= \{x \mid x \in A\} && \text{by absorption law of logic} \\
&= A && \text{by meaning of set-builder notation.}
\end{aligned}
$$

Proof of the Absorption Law

Absorption Law: $\boxed{A \cup (A \cap B) = A}$

$$\begin{aligned}
A \cup (A \cap B) &= \{x \mid x \in A \cup (A \cap B)\} && \text{by meaning of set-builder notation} \\
&= \{x \mid x \in A \vee x \in (A \cap B)\} && \text{by definition of union} \\
&= \{x \mid x \in A \vee (x \in A \wedge x \in B)\} && \text{by definition of intersection} \\
&= \{x \mid x \in A\} && \text{by absorption law of logic} \\
&= A && \text{by meaning of set-builder notation.}
\end{aligned}$$

Proof of the Absorption Law

Absorption Law: $\boxed{A \cup (A \cap B) = A}$

$$
\begin{aligned}
A \cup (A \cap B) &= \{x \mid x \in A \cup (A \cap B)\} && \text{by meaning of set-builder notation} \\
&= \{x \mid x \in A \vee x \in (A \cap B)\} && \text{by definition of union} \\
&= \{x \mid x \in A \vee (x \in A \wedge x \in B)\} && \text{by definition of intersection} \\
&= \{x \mid x \in A\} && \text{by absorption law of logic} \\
&= A && \text{by meaning of set-builder notation.}
\end{aligned}
$$

Proof of the Absorption Law

Absorption Law: $\boxed{A \cup (A \cap B) = A}$

$$
\begin{aligned}
A \cup (A \cap B) &= \{x \mid x \in A \cup (A \cap B)\} && \text{by meaning of set-builder notation} \\
&= \{x \mid x \in A \vee x \in (A \cap B)\} && \text{by definition of union} \\
&= \{x \mid x \in A \vee (x \in A \wedge x \in B)\} && \text{by definition of intersection} \\
&= \{x \mid x \in A\} && \text{by absorption law of logic} \\
&= A && \text{by meaning of set-builder notation.}
\end{aligned}
$$

Proof of the Absorption Law

Absorption Law: $\boxed{A \cup (A \cap B) = A}$

$$
\begin{aligned}
A \cup (A \cap B) &= \{x \mid x \in A \cup (A \cap B)\} && \text{by meaning of set-builder notation} \\
&= \{x \mid x \in A \vee x \in (A \cap B)\} && \text{by definition of union} \\
&= \{x \mid x \in A \vee (x \in A \wedge x \in B)\} && \text{by definition of intersection} \\
&= \{x \mid x \in A\} && \text{by absorption law of logic} \\
&= A && \text{by meaning of set-builder notation.}
\end{aligned}
$$

Proof of the Distributive Law

Distributive Law: $\boxed{A \cup (B \cap C) = (A \cup B) \cap (A \cup C)}$

$$
\begin{aligned}
A \cup (B \cap C) &= \{x \mid x \in A \cup (B \cap C)\} && \text{by meaning of set-builder notation} \\
&= \{x \mid x \in A \vee x \in (B \cap C)\} && \text{by definition of union} \\
&= \{x \mid x \in A \vee (x \in B \wedge x \in C)\} && \text{by definition of intersection} \\
&= \{x \mid (x \in A \vee x \in B) \wedge (x \in A \vee x \in C)\} && \text{by distributive law for logic} \\
&= \{x \mid x \in (A \cup B) \wedge x \in (A \cup C)\} && \text{by definition of union} \\
&= \{x \mid x \in (A \cup B) \cap (A \cup C)\} && \text{by definition of intersection} \\
&= (A \cup B) \cap (A \cup C) && \text{by meaning of set-builder notation.}
\end{aligned}
$$

Proof of the Distributive Law

Distributive Law: $\boxed{A \cup (B \cap C) = (A \cup B) \cap (A \cup C)}$

$$
\begin{aligned}
A \cup (B \cap C) &= \{x \mid x \in A \cup (B \cap C)\} && \text{by meaning of set-builder notation} \\
&= \{x \mid x \in A \vee x \in (B \cap C)\} && \text{by definition of union} \\
&= \{x \mid x \in A \vee (x \in B \wedge x \in C)\} && \text{by definition of intersection} \\
&= \{x \mid (x \in A \vee x \in B) \wedge (x \in A \vee x \in C)\} && \text{by distributive law for logic} \\
&= \{x \mid x \in (A \cup B) \wedge x \in (A \cup C)\} && \text{by definition of union} \\
&= \{x \mid x \in (A \cup B) \cap (A \cup C)\} && \text{by definition of intersection} \\
&= (A \cup B) \cap (A \cup C) && \text{by meaning of set-builder notation.}
\end{aligned}
$$

Proof of the Distributive Law

Distributive Law: $\boxed{A \cup (B \cap C) = (A \cup B) \cap (A \cup C)}$

$$
\begin{aligned}
A \cup (B \cap C) &= \{x \mid x \in A \cup (B \cap C)\} && \text{by meaning of set-builder notation} \\
&= \{x \mid x \in A \lor x \in (B \cap C)\} && \text{by definition of union} \\
&= \{x \mid x \in A \lor (x \in B \land x \in C)\} && \text{by definition of intersection} \\
&= \{x \mid (x \in A \lor x \in B) \land (x \in A \lor x \in C)\} && \text{by distributive law for logic} \\
&= \{x \mid x \in (A \cup B) \land x \in (A \cup C)\} && \text{by definition of union} \\
&= \{x \mid x \in (A \cup B) \cap (A \cup C)\} && \text{by definition of intersection} \\
&= (A \cup B) \cap (A \cup C) && \text{by meaning of set-builder notation.}
\end{aligned}
$$

Proof of the Distributive Law

Distributive Law: $\boxed{A \cup (B \cap C) = (A \cup B) \cap (A \cup C)}$

$$
\begin{aligned}
A \cup (B \cap C) &= \{x \mid x \in A \cup (B \cap C)\} && \text{by meaning of set-builder notation} \\
&= \{x \mid x \in A \lor x \in (B \cap C)\} && \text{by definition of union} \\
&= \{x \mid x \in A \lor (x \in B \land x \in C)\} && \text{by definition of intersection} \\
&= \{x \mid (x \in A \lor x \in B) \land (x \in A \lor x \in C)\} && \text{by distributive law for logic} \\
&= \{x \mid x \in (A \cup B) \land x \in (A \cup C)\} && \text{by definition of union} \\
&= \{x \mid x \in (A \cup B) \cap (A \cup C)\} && \text{by definition of intersection} \\
&= (A \cup B) \cap (A \cup C) && \text{by meaning of set-builder notation.}
\end{aligned}
$$

Proof of the Distributive Law

Distributive Law: $\boxed{A \cup (B \cap C) = (A \cup B) \cap (A \cup C)}$

$$
\begin{aligned}
A \cup (B \cap C) &= \{x \mid x \in A \cup (B \cap C)\} && \text{by meaning of set-builder notation} \\
&= \{x \mid x \in A \vee x \in (B \cap C)\} && \text{by definition of union} \\
&= \{x \mid x \in A \vee (x \in B \wedge x \in C)\} && \text{by definition of intersection} \\
&= \{x \mid (x \in A \vee x \in B) \wedge (x \in A \vee x \in C)\} && \text{by distributive law for logic} \\
&= \{x \mid x \in (A \cup B) \wedge x \in (A \cup C)\} && \text{by definition of union} \\
&= \{x \mid x \in (A \cup B) \cap (A \cup C)\} && \text{by definition of intersection} \\
&= (A \cup B) \cap (A \cup C) && \text{by meaning of set-builder notation.}
\end{aligned}
$$

Proof of the Distributive Law

Distributive Law: $\boxed{A \cup (B \cap C) = (A \cup B) \cap (A \cup C)}$

$$
\begin{aligned}
A \cup (B \cap C) &= \{x \mid x \in A \cup (B \cap C)\} && \text{by meaning of set-builder notation} \\
&= \{x \mid x \in A \lor x \in (B \cap C)\} && \text{by definition of union} \\
&= \{x \mid x \in A \lor (x \in B \land x \in C)\} && \text{by definition of intersection} \\
&= \{x \mid (x \in A \lor x \in B) \land (x \in A \lor x \in C)\} && \text{by distributive law for logic} \\
&= \{x \mid x \in (A \cup B) \land x \in (A \cup C)\} && \text{by definition of union} \\
&= \{x \mid x \in (A \cup B) \cap (A \cup C)\} && \text{by definition of intersection} \\
&= (A \cup B) \cap (A \cup C) && \text{by meaning of set-builder notation.}
\end{aligned}
$$

Proof of the Distributive Law

Distributive Law: $\boxed{A \cup (B \cap C) = (A \cup B) \cap (A \cup C)}$

$$\begin{aligned}
A \cup (B \cap C) &= \{x \mid x \in A \cup (B \cap C)\} && \text{by meaning of set-builder notation} \\
&= \{x \mid x \in A \lor x \in (B \cap C)\} && \text{by definition of union} \\
&= \{x \mid x \in A \lor (x \in B \land x \in C)\} && \text{by definition of intersection} \\
&= \{x \mid (x \in A \lor x \in B) \land (x \in A \lor x \in C)\} && \text{by distributive law for logic} \\
&= \{x \mid x \in (A \cup B) \land x \in (A \cup C)\} && \text{by definition of union} \\
&= \{x \mid x \in (A \cup B) \cap (A \cup C)\} && \text{by definition of intersection} \\
&= (A \cup B) \cap (A \cup C) && \text{by meaning of set-builder notation.}
\end{aligned}$$

Set Identity: Difference of Sets

Problem

Let A, B, and C be sets. Show that $(A - B) - C = (A - C) - (B - C)$.

Proof: We prove equality by showing that an arbitrary element x belongs to the left-hand side if and only if it belongs to the right-hand side.

$$\begin{aligned} x \in (A - B) - C &\iff x \in (A - B) \;\wedge\; x \notin C \\ &\iff (x \in A \wedge x \notin B) \wedge x \notin C \\ &\iff x \in A \wedge x \notin C \wedge x \notin B \\ &\iff x \in (A - C) \wedge x \notin (B - C) \\ &\iff x \in (A - C) - (B - C). \end{aligned}$$

Set Identity: Difference of Sets

Problem

Let A, B, and C be sets. Show that $(A - B) - C = (A - C) - (B - C)$.

Proof: We prove equality by showing that an arbitrary element x belongs to the left-hand side if and only if it belongs to the right-hand side.

$$\begin{aligned}
x \in (A - B) - C &\iff x \in (A - B) \ \land \ x \notin C \\
&\iff (x \in A \land x \notin B) \land x \notin C \\
&\iff x \in A \land x \notin C \land x \notin B \\
&\iff x \in (A - C) \land x \notin (B - C) \\
&\iff x \in (A - C) - (B - C).
\end{aligned}$$

Set Identity: Subsets and Complements

Problem

Let A and B be subsets of a universal set U. Show that $A \subseteq B \iff \overline{B} \subseteq \overline{A}$.

($\Rightarrow$) Suppose $A \subseteq B$. We want to show that $\overline{B} \subseteq \overline{A}$.

Let $x \in \overline{B}$. Then $x \notin B$. Since $A \subseteq B$, $x \notin A$, and hence $x \in \overline{A}$. Thus, $\overline{B} \subseteq \overline{A}$.

($\Leftarrow$) Suppose $\overline{B} \subseteq \overline{A}$. Let $x \in A$. We want to show that $x \in B$.

Suppose, for contradiction, that $x \notin B$. Then $x \in \overline{B}$. Since $\overline{B} \subseteq \overline{A}$, we obtain $x \in \overline{A}$, which means $x \notin A$ — a contradiction. Therefore, $x \in B$, and hence $A \subseteq B$.

Set Identity: Subsets and Complements

Problem

Let A and B be subsets of a universal set U. Show that $A \subseteq B \iff \overline{B} \subseteq \overline{A}$.

(⇒) Suppose $A \subseteq B$. We want to show that $\overline{B} \subseteq \overline{A}$.

Let $x \in \overline{B}$. Then $x \notin B$. Since $A \subseteq B$, $x \notin A$, and hence $x \in \overline{A}$. Thus, $\overline{B} \subseteq \overline{A}$.

(⇐) Suppose $\overline{B} \subseteq \overline{A}$. Let $x \in A$. We want to show that $x \in B$.

Suppose, for contradiction, that $x \notin B$. Then $x \in \overline{B}$. Since $\overline{B} \subseteq \overline{A}$, we obtain $x \in \overline{A}$, which means $x \notin A$ — a contradiction. Therefore, $x \in B$, and hence $A \subseteq B$.

Set Identity: Subsets and Complements

Problem

Let A and B be subsets of a universal set U. Show that $A \subseteq B \iff \overline{B} \subseteq \overline{A}$.

($\Rightarrow$) Suppose $A \subseteq B$. We want to show that $\overline{B} \subseteq \overline{A}$.

Let $x \in \overline{B}$. Then $x \notin B$. Since $A \subseteq B$, $x \notin A$, and hence $x \in \overline{A}$. Thus, $\overline{B} \subseteq \overline{A}$.

($\Leftarrow$) Suppose $\overline{B} \subseteq \overline{A}$. Let $x \in A$. We want to show that $x \in B$.

Suppose, for contradiction, that $x \notin B$. Then $x \in \overline{B}$. Since $\overline{B} \subseteq \overline{A}$, we obtain $x \in \overline{A}$, which means $x \notin A$ — a contradiction. Therefore, $x \in B$, and hence $A \subseteq B$.

Set Identity: Cartesian Product and Difference

Problem

Let A, B, and C be sets. Show that $A \times (B - C) = (A \times B) - (A \times C)$.

($\subseteq$) Let $(x, y) \in A \times (B - C)$. Then $x \in A$ and $y \in B - C$. Hence $y \in B$ and $y \notin C$. Therefore, $(x, y) \in A \times B$ but $(x, y) \notin A \times C$. Thus $(x, y) \in (A \times B) - (A \times C)$. Hence, $A \times (B - C) \subseteq (A \times B) - (A \times C)$.

($\supseteq$) Let $(x, y) \in (A \times B) - (A \times C)$. Then $(x, y) \in A \times B$ and $(x, y) \notin A \times C$. Thus $x \in A$ and $y \in B$.

Since $x \in A$ and $(x, y) \notin A \times C$, we must have $y \notin C$. Hence $y \in B - C$, and therefore $(x, y) \in A \times (B - C)$.

Thus, $\boxed{(A \times B) - (A \times C) \subseteq A \times (B - C).}$

Set Identity: Cartesian Product and Difference

Problem

Let A, B, and C be sets. Show that $A \times (B - C) = (A \times B) - (A \times C)$.

($\subseteq$) Let $(x, y) \in A \times (B - C)$. Then $x \in A$ and $y \in B - C$. Hence $y \in B$ and $y \notin C$. Therefore, $(x, y) \in A \times B$ but $(x, y) \notin A \times C$. Thus $(x, y) \in (A \times B) - (A \times C)$. Hence, $A \times (B - C) \subseteq (A \times B) - (A \times C)$.

($\supseteq$) Let $(x, y) \in (A \times B) - (A \times C)$. Then $(x, y) \in A \times B$ and $(x, y) \notin A \times C$. Thus $x \in A$ and $y \in B$.

Since $x \in A$ and $(x, y) \notin A \times C$, we must have $y \notin C$. Hence $y \in B - C$, and therefore $(x, y) \in A \times (B - C)$.

Thus, $\boxed{(A \times B) - (A \times C) \subseteq A \times (B - C).}$

Set Identity: Cartesian Product and Difference

Problem

Let A, B, and C be sets. Show that $A \times (B - C) = (A \times B) - (A \times C)$.

($\subseteq$) Let $(x, y) \in A \times (B - C)$. Then $x \in A$ and $y \in B - C$. Hence $y \in B$ and $y \notin C$. Therefore, $(x, y) \in A \times B$ but $(x, y) \notin A \times C$. Thus $(x, y) \in (A \times B) - (A \times C)$. Hence, $A \times (B - C) \subseteq (A \times B) - (A \times C)$.

($\supseteq$) Let $(x, y) \in (A \times B) - (A \times C)$. Then $(x, y) \in A \times B$ and $(x, y) \notin A \times C$. Thus $x \in A$ and $y \in B$.

Since $x \in A$ and $(x, y) \notin A \times C$, we must have $y \notin C$. Hence $y \in B - C$, and therefore $(x, y) \in A \times (B - C)$.

Thus, $\boxed{(A \times B) - (A \times C) \subseteq A \times (B - C).}$

Set Identities: Practice

$\mathcal{Q}$. **Show that**

$$A - B = A \cap \overline{B}.$$

Set Identities: Practice

$\mathcal{Q}$. **Show that**

$$A - B = A \cap \overline{B}.$$

$\mathcal{A}$. $x \in (A - B) \iff x \in A \wedge x \notin B \iff x \in A \wedge x \in \overline{B} \iff x \in A \cap \overline{B}$

Set Identities: Practice

$\mathcal{Q}$. **Show that**

$$A - B = A \cap \overline{B}.$$

$\mathcal{A}$. $x \in (A - B) \iff x \in A \wedge x \notin B \iff x \in A \wedge x \in \overline{B} \iff x \in A \cap \overline{B}$

$\mathcal{Q}$. **Show that**

$$(A \cap B) \cup (A \cap \overline{B}) = A.$$

Set Identities: Practice

$\mathcal{Q}$. **Show that**

$$A - B = A \cap \overline{B}.$$

$\mathcal{A}$. $x \in (A - B) \iff x \in A \land x \notin B \iff x \in A \land x \in \overline{B} \iff x \in A \cap \overline{B}$

$\mathcal{Q}$. **Show that**

$$(A \cap B) \cup (A \cap \overline{B}) = A.$$

$\mathcal{A}$. Using the distributive law,

$$\begin{aligned}(A \cap B) \cup (A \cap \overline{B}) &= A \cap (B \cup \overline{B}) \\ &= A \cap U \\ &= A.\end{aligned}$$

Quick Check: Set Operations

What can you say about the sets A and B if we know that

$\mathcal{Q}$. $A \cup B = A$?

Quick Check: Set Operations

What can you say about the sets A and B if we know that

$\mathcal{Q}$. $A \cup B = A$?

$\mathcal{A}$. $B \subseteq A$.

Quick Check: Set Operations

What can you say about the sets A and B if we know that

$\mathfrak{Q}$. $A \cup B = A$?

$\mathcal{A}$. $B \subseteq A$.

$\mathfrak{Q}$. What can you say if $A \cap B = A$?

Quick Check: Set Operations

What can you say about the sets A and B if we know that

$\mathcal{Q}$. $A \cup B = A$?

$\mathcal{A}$. $B \subseteq A$.

$\mathcal{Q}$. What can you say if $A \cap B = A$?

$\mathcal{A}$. $A \subseteq B$.

Quick Check: Set Operations

What can you say about the sets A and B if we know that

$\mathfrak{Q}$. $A \cup B = A$?

$\mathcal{A}$. $B \subseteq A$.

$\mathfrak{Q}$. What can you say if $A \cap B = A$?

$\mathcal{A}$. $A \subseteq B$.

$\mathfrak{Q}$. What can you say if $A - B = A$?

Quick Check: Set Operations

What can you say about the sets A and B if we know that

$\mathfrak{Q}$. $A \cup B = A$?

$\mathcal{A}$. $B \subseteq A$.

$\mathfrak{Q}$. What can you say if $A \cap B = A$?

$\mathcal{A}$. $A \subseteq B$.

$\mathfrak{Q}$. What can you say if $A - B = A$?

$\mathcal{A}$. $A \cap B = \emptyset$.

Quick Check: Set Operations

What can you say about the sets A and B if we know that

$\mathcal{Q}$. $A \cup B = A$?

$\mathcal{A}$. $B \subseteq A$.

$\mathcal{Q}$. What can you say if $A \cap B = A$?

$\mathcal{A}$. $A \subseteq B$.

$\mathcal{Q}$. What can you say if $A - B = A$?

$\mathcal{A}$. $A \cap B = \emptyset$.

$\mathcal{Q}$. What can you say if $A \cap B = B \cap A$?

Quick Check: Set Operations

What can you say about the sets A and B if we know that

$\mathfrak{Q}$. $A \cup B = A$?

$\mathcal{A}$. $B \subseteq A$.

$\mathfrak{Q}$. What can you say if $A \cap B = A$?

$\mathcal{A}$. $A \subseteq B$.

$\mathfrak{Q}$. What can you say if $A - B = A$?

$\mathcal{A}$. $A \cap B = \emptyset$.

$\mathfrak{Q}$. What can you say if $A \cap B = B \cap A$?

$\mathcal{A}$. Nothing special. This is always true because intersection is commutative.

Quick Check: Set Operations

What can you say about the sets A and B if we know that

$\mathfrak{Q}$. $A \cup B = A$?

$\mathcal{A}$. $B \subseteq A$.

$\mathfrak{Q}$. What can you say if $A \cap B = A$?

$\mathcal{A}$. $A \subseteq B$.

$\mathfrak{Q}$. What can you say if $A - B = A$?

$\mathcal{A}$. $A \cap B = \emptyset$.

$\mathfrak{Q}$. What can you say if $A \cap B = B \cap A$?

$\mathcal{A}$. Nothing special. This is always true because intersection is commutative.

$\mathfrak{Q}$. What can you say if $A - B = B - A$?

Quick Check: Set Operations

What can you say about the sets A and B if we know that

$\mathfrak{Q}$. $A \cup B = A$?

$\mathcal{A}$. $B \subseteq A$.

$\mathfrak{Q}$. What can you say if $A \cap B = A$?

$\mathcal{A}$. $A \subseteq B$.

$\mathfrak{Q}$. What can you say if $A - B = A$?

$\mathcal{A}$. $A \cap B = \emptyset$.

$\mathfrak{Q}$. What can you say if $A \cap B = B \cap A$?

$\mathcal{A}$. Nothing special. This is always true because intersection is commutative.

$\mathfrak{Q}$. What can you say if $A - B = B - A$?

$\mathcal{A}$. $A = B$.

Thank You for your attention!

Any questions?